

Slide 48 – Variance Inflation Factor (VIF)

What problem is VIF solving?

Suppose you're predicting house prices.

Features:

House Size
Number of Rooms
Square Feet
Built-up Area

Notice:

House Size $\approx$ Square Feet $\approx$ Built-up Area

These variables are almost telling the model the same thing.

This is called:

Multicollinearity

VIF is used to detect this.

Simple Analogy

Imagine 5 people in a meeting.

You ask:

How much traffic is there?

All 5 repeat:

Heavy traffic
Heavy traffic
Heavy traffic
Heavy traffic
Heavy traffic

You're not getting new information.

Same thing happens with highly correlated features.

VIF Interpretation

VIF	Meaning
-----	---------

1	No correlation
---	----------------

1-5	Moderate correlation
-----	----------------------

>5	High correlation
----	------------------

>10	Serious issue
-----	---------------

Slide 49 – Correlation Matrix

A Correlation Matrix shows how strongly variables are related.

Values range from:

-1 to +1

Value	Meaning
-------	---------

+1	Perfect Positive Correlation
----	------------------------------

-1	Perfect Negative Correlation
----	------------------------------

0	No Correlation
---	----------------

Example

Feature A	Feature B	Correlation
-----------	-----------	-------------

House Size	Built-up Area	0.95
------------	---------------	------

Salary	Experience	0.8
--------	------------	-----

Height	House Price	0.02
--------	-------------	------

Interpretation:

0.95 = Almost identical

0.8 = Strong relationship

0.02 = No relationship

Slide 50 – Feature Engineering

Feature Engineering means:

Improving input features before training

Examples:

Raw Data

DOB = 12-01-1990

Convert to:

Age = 36

Better feature.

Another Example

Raw:

Salary = 50000

Experience = 5

New Feature:

Salary per Year Experience

Sometimes a new feature is more useful than the originals.

Slide 51 – Feature Selection

Feature Selection means:

Keep useful columns

Remove useless columns

Example

Employee Salary Prediction

Features:

Age

Experience

Education

Salary

Employee ID

Passport Number

Useful:

Age

Experience
Education

Remove:

Employee ID
Passport Number

because they don't help prediction.

Why?

Benefits:

Faster Training
Less Overfitting
Better Accuracy
Simpler Model

Slide 52 – High Correlation vs Low Correlation

Imagine:

High Correlation

House Size
Built-up Area

Both tell similar stories.

Correlation:

0.95

Low Correlation

House Size
Owner Mobile Number

Correlation:

0.01

Mobile number has nothing to do with price.

Remove it.

Slide 53 – Why Feature Selection Matters

Without feature selection:

100 Features

Only:

10 Useful

90 Noise

Model gets confused.

With feature selection:

Keep 10 useful features

Model becomes:

Faster

More accurate

More interpretable

Slide 54 – Model Selection

Question:

Which model should I use?

Choices:

Linear Regression

Ridge

Lasso

Elastic Net

Decision Tree

Random Forest

Model Selection means:

Try multiple models

Choose best performer

Slide 55 – Comparing Models

Suppose:

Model	RMSE
Linear Regression	15
Ridge	12
Lasso	10

Choose:

Lasso

because:

Lower RMSE = Better

Slide 56 – Simplicity vs Accuracy

Important concept.

Model A

Accuracy = 92%
Features = 5

Model B

Accuracy = 93%
Features = 100

Which is better?

Often:

Model A

because it's easier to understand and maintain.

Slide 57 – Underfitting vs Overfitting Review

Underfitting

Too Simple

Example:

Straight line for curved data

Result:

Poor Training Accuracy
Poor Test Accuracy

Overfitting

Too Complex

Result:

Excellent Training Accuracy
Poor Test Accuracy

Good Fit

Balanced

Result:

Good Training Accuracy
Good Test Accuracy

Slide 58 – Bias vs Variance

High Bias

Model assumes too much.

Example:

Straight line everywhere

Result:

Underfitting

High Variance

Model memorizes data.

Result:

Overfitting

Goal:

Low Bias
Low Variance

Slide 59 – Putting Everything Together

Regression Workflow:

```
Collect Data
  ↓
Clean Data
  ↓
Feature Engineering
  ↓
Feature Selection
  ↓
Train Model
  ↓
Evaluate Model
  ↓
Apply Ridge/Lasso if needed
  ↓
Optimize using Gradient Descent
  ↓
Deploy
```

Slide 60 – Key Takeaways

Regression

Predicts continuous values.

Examples:

House Price
Temperature
Sales
Revenue

Evaluation Metrics

MAE
MSE
RMSE
 R^2

Regularization

Ridge → Shrinks coefficients

Lasso → Shrinks + Feature Selection

Elastic Net → Combination of both

Gradient Descent

Finds the best values of m and c
by minimizing error

Multicollinearity

Use:

VIF
Correlation Matrix

to detect correlated features.

Interview Memory Sheet

Regression = Predict Numbers

MAE = Average Error

MSE = Squared Error

RMSE = Error in original units

R^2 = How well model explains data

Ridge = L2 = Shrinks

Lasso = L1 = Shrinks + Selects

Elastic Net = Ridge + Lasso

Gradient Descent = Finds Best Line

VIF = Detect Multicollinearity

Feature Engineering = Create Better Features

Feature Selection = Keep Useful Features

If you're preparing for interviews, these slides (42–60) together form one complete story:
"Build a regression model → measure error → avoid overfitting → optimize it → remove bad features → select the best model." That's the mental model to remember.

